

## CURRICULUM VITAE

**NAME:** Sandeep  
**YEAR OF BIRTH:** 1990  
**PROFESSION:** Mechanical Engineer  
**NATIONALITY:** Indian  
**CONTACT:** [sandeep\\_ps@thapar.edu](mailto:sandeep_ps@thapar.edu)  
[sandeepnain.kuk@gmail.com](mailto:sandeepnain.kuk@gmail.com)  
**Mobile no. -** +919416703919

### Professional Online Profile:

**Google Scholar:** [Sandeep Nain - Google Scholar](#)  
**ORCID:** <https://orcid.org/0000-0003-1802-6797>  
**Research Gate:** <https://www.researchgate.net/profile/sandeep-Nain>  
**LinkedIn:** <https://www.linkedin.com/in/sandeep-nain-ph-d-28406813a>  
**Scopus:** [Nain, Sandeep - Author details - Scopus](#)

**SPECIALIZATION:** Thermo-fluid transport processes, biophysics, and biomedical applications using computational fluid dynamics. Modelling of biological systems based on morphology and hemodynamics. Ai-Driven Cancer detection and segmentation.

**CURRENT POST:** Research Associate, CoE-DSAI, TIET-UQ joint research project in **Thapar Institute of Engineering & Technology (TIET)**, Patiala, Punjab, India.

---

## EDUCATION

|                                                                                                                                                         |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>CoE-DSAI, TIET-UQ, TIET, Patiala, Punjab, India</b><br><b>Post Doctorate</b> in data science and AI inclusion in cancer and CFD research.            | <b>2025-present</b> |
| <b>Thapar Institute of Engineering and Technology, Punjab, India</b><br><b>Doctor of Philosophy</b> in Mechanical Engineering.                          | <b>2018-2024</b>    |
| <b>University Institute of Engineering and Technology, Kurukshetra University Kurukshetra, India</b><br><b>M. Tech.</b> in Thermal Engineering, 8.28/10 | <b>2015-2017</b>    |
| <b>Chaudhary Devilal University, Sirsa, Haryana, India</b><br><b>B. Tech</b> in Mechanical Engineering, 68.3%                                           | <b>2008-2011</b>    |
| <b>Government Polytechnic Hisar, Hisar, Haryana, India</b><br><b>Diploma</b> in Mechanical Engineering, 71.94%                                          | <b>2005-2008</b>    |

---

## RESEARCH EXPERIENCE

|                                                                                                                                                                                                                                                                                                                 |                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>TIET, Punjab, India</b>                                                                                                                                                                                                                                                                                      |                                                          |
| <ul style="list-style-type: none"><li><b>Research Associate (RA)</b>, Center of Excellence in Data Science and Artificial Intelligence (CoE-DSAI), <b>TIET-UQ</b>, “AI-Driven Modelling and Optimization of Breast Cancer Detection and Treatment through Magnetically Induced Hyperthermia Therapy”.</li></ul> | <b>July 2025</b><br><b>Present</b><br><b>Sept. 2023-</b> |
| <ul style="list-style-type: none"><li><b>Senior Research Fellow (SRF)</b>, TIET-VT (Thapar-Virginia Tech) Centre of Excellence for Emerging Materials (<b>CEEMS</b>), “Computational and Experimental Investigation for Optimizing the Magnetic Nano-Particles Hyperthermia”.</li></ul>                         | <b>June 2024</b>                                         |

- 
- **Junior Research Fellow (JRF)** in TIET-VT (Thapar-Virginia Tech) **CEEMS**,  
“Computational and Experimental Investigation for Optimizing the Magnetic Nano-  
Particles Hyperthermia”. **Sept. 2021-  
August 2023**

**Research projects @ collaborations:**

1. Computational and Experimental Investigation for Optimizing the Magnetic Nano-Particles Hyperthermia @ TIET-VT (Thapar-Virginia Tech) **CEEMS. (Finished)**
  2. AI-Driven Breast Cancer Detection, segmentation and Treatment through Magnetically Induced Hyperthermia. @ CoE-DSAI), **TIET-University of Queensland. (Ongoing)**
  3. Environmental pollution modelling using AI4PDEs solver based on CNN architecture @ TIET-**Imperial college London. (Ongoing).**
- **Thapar Institute of Engineering and Technology, Punjab, India**  
**Regular Ph.D. scholar** in the Mechanical Engineering Department (MED), **Jan 2018 -  
June 2024**
- 

**LIST OF PUBLICATIONS**

**Peer-Reviewed Published Journal Articles**

1. **Nain, Sandeep**, Neeraj Kumar, and Pramod Kumar Avti. "Computational investigation of the tumor position and ambient conditions on magnetic nanoparticle thermo-therapy." *Thermal Science and Engineering Progress* 34 (2022): 101396. <https://doi.org/10.1016/j.tsep.2022.101396> (SCI Journal, IF - 5.4)
2. **Nain, Sandeep**, Neeraj Kumar, Bhupendra Chudasama, and Pramod Kumar Avti. "The SLP estimation of the nanoparticle systems using size-dependent magnetic properties for the magnetic hyperthermia therapy." *Journal of Magnetism and Magnetic Materials* 565 (2023): 170219. <https://doi.org/10.1016/j.jmmm.2022.170219> (SCI Journal, IF - 3).
3. **S. Nain**, N. Kumar, and P. K. Avti, “Tumor size dependent MNP dose evaluation in realistic breast tumor models for effective magnetic hyperthermia.,” *Med. Eng. Phys.*, p. 104065, 2023, <https://doi.org/10.1016/j.medengphy.2023.104065> (SCI Journal, IF - 2.3).
4. **Nain, S.**, Kumar, N., Avti, P.K., 2025. Effect of Arterial Blood Flow on Magnetic Nanoparticle Thermo-therapy Applied on a Realistic Breast Tumor Model. *Int j numer method biomed eng* 41, e70039. <https://doi.org/10.1002/cnm.70039> (SCI Journal, IF - 2.4).

**Peer-Reviewed Journal Articles (Under review)**

1. **Nain, Sandeep**, Neeraj Kumar, and Pramod Kumar Avti “Magnetic hyperthermia and its dependency on nanoparticles, tissues, and other critical parameters: a comparative study” in *Biomedical Physics & Engineering Express*.

**Ready for Communication Articles**

1. **S. Nain**, N. Kumar, and P. K. Avti, “CFD-based Design and Performance analysis of digital twin for cancer prediction and treatment using magnetic fluid hyperthermia.

#### **Articles in National and International Conferences**

- **S. Nain**, N. Kumar, and P. K. Avti. (2024). Oral paper presented in the “**International Conference on Sustainable Development in Chemical and Environmental Engineering**” 2024 in *TIET, Patiala, India*, entitled "Computational evaluation of optimum tumor size-dependent MNP dose for realistic breast tumor models with different injection strategies in magnetic hyperthermia".

#### **TEACHING EXPERIENCE**

---

|                                                                                                                          |                             |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Teaching Assistantship</b> of 5-7 hours as per the UGC norms during the period of JRF and SRF in CEEMS, TIET, Patiala | <b>Sept. 2021-June 2024</b> |
| <b>Teaching Associate</b> in MED, Thapar Institute of Engineering and Technology, Punjab, India                          | <b>July 2018- June 2021</b> |
| <b>Lecturer</b> in karnal institute of technology and management (KITM), Karnal, Haryana, India                          | <b>May 2012- June 2013</b>  |

#### **Teaching responsibilities:**

- Delivered individually the quality education for subjects: Engineering Drawing (Semester -1<sup>st</sup>), Machine Drawing (4<sup>th</sup> Semester.), Auto cad class (2<sup>nd</sup> Sem.)
  - Effective engagement of different Labs i.e., Kinematics of Machine (3<sup>rd</sup>), Fluid Machines, IC engines (4<sup>th</sup>), Heat and Mass Transfer (6<sup>th</sup>), Refrigeration and Air Conditioning (8<sup>th</sup>), Automobile Engineering Lab (Semester-8<sup>th</sup>) was done with excellent management skills, effective communication, and innovative teaching methods.
  - Assigned class tutorials on subjects, i.e., thermodynamics, FM, HMT, Power Plant, and Solar Engineering, which were handled with excellent classroom management and teaching skills.
  - Provided assistance to senior faculty in various routine academic activities in MED.
  - Performed invigilation duties in mid-term and end-term examinations.
  - Maintained records of continuous assessment of the assigned subjects.
- 

#### **EXPERIENCE OF STUDENTS SUPERVISION**

1. Supervised a Master’s student project on “**The effect of convection phenomenon during magnetic nanoparticle hyperthermia treatment.**”
2. Supervised a Master’s student project on “**Investigation of Microchannel Heat sink and Graphene-Based heat spreaders for thermal management of 3D TSV chip.**”

#### **CONFERENCES/ WORKSHOPS/ SEMINARS**

- **International Conference on Sustainable Development in Chemical and Environmental Engineering (SDCEE-2024);** at TIET, Patiala, India.
- **Faculty Development Program on “computational fluid dynamics”** at UIET, KUK, in collaboration with IIT Bombay.

#### **ACHIEVEMENTS, HONORS, AND EXTRACURRICULAR ACTIVITIES**

|                                                                                                                                                                  |                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Funded for JRF and SRF</b> , in Centre of Excellence for Emerging Materials, TIET-VT Patiala, funded research project                                         | <b>2021-2024</b>            |
| <b>Funded for the PhD</b> , Mechanical Engineering Department, TIET, Patiala                                                                                     | <b>2018-2021</b>            |
| <b>Fellowship during M.Tech</b> , from AICTE, India.                                                                                                             | <b>2015-2017</b>            |
| <b>Graduate Aptitude Test of Engineering (GATE) QUALIFIED</b> with a best GATE (2015) score of <b>454 /1000</b> and All India Rank 15753 out of 185758 students. | <b>2014, 2015, and 2017</b> |
| <b>Diploma in German Language</b> , Kurukshetra University Kurukshetra, India                                                                                    | <b>2018</b>                 |
| <b>Certificate Course in German Language</b> , Kurukshetra University Kurukshetra, India                                                                         | <b>2017</b>                 |
| <b>Gold medal</b> in Discus throw in Thapar Institute Olympic games.                                                                                             | <b>2018</b>                 |
| <b>Silver medal</b> in Discus throw in Inter-Tech University games.                                                                                              | <b>2019</b>                 |
| <b>2<sup>nd</sup> runner up</b> in 10 km running organized by Patiala Health Club, Patiala.                                                                      | <b>2019</b>                 |
| <b>Participate</b> in a six-day <b>Faculty Development Programme</b> on “computational fluid dynamics” at UIET, KUK, in collaboration with IIT Bombay.           | <b>Feb 2017</b>             |
| <b>National Cadet Corps (NCC) “C”</b> certificate from Ministry of Defense, govt. of India.                                                                      | <b>2009</b>                 |

#### **Ph.D. RESEARCH TOPIC**

- “Computational and Experimental Investigation for Optimizing the Magnetic Nanoparticles Hyperthermia”.

#### **ME-THESIS RESEARCH TOPIC**

- “Computational Modeling and Analysis of 2d Unsteady Heat Convection in Slug Flow Using Physical Law based FVM”.

#### **TRAINING AND PROJECTS**

- Four weeks of training in “**Escorts Limited**”, Agri Machinery Group in ‘**gear and Shaft Area**’ (July 2007).
- Six-week Training in “**Yamaha Motor Pvt. India Limited**” in ‘**Gear manufacturing Area**’ (June 2009).
- Physical Modeling of “**Frictional Energy Conversion from railway track**” as a minor project in B. Tech.
- Physical Modeling of “**Over-Head Crane**” as a major project in B. Tech.
- Computational Modeling of convection **Heat Transfer in slug flow** in UIET, KUK during dissertation Work in M. Tech Program.
- Development of solver using **Finite volume method for bio-heat** application in hyperthermia therapy.
- Computationally investigating the therapeutic effect of different injection strategies in real patient data.
- Experimentally evaluation of the temperature distribution by the **different MNP distributions in a gel phantom**.

## **SOFTWARE and PROGRAMMING SKILLS**

- **MATLAB, SCI-LAB**
- **COMSOL MULTI-PHYSICS, ANSYS**
- **SOLID WORK, AUTO-CAD, G-MESH, ORIGIN**
- **Fortran, C++, PYTHON, R, and ML (basics)**

## **REFERENCES**

### **Dr. Neeraj Kumar,**

Assistant Professor,

Mechanical Engineering Department,

Thapar Institute of Engineering and Technology, Patiala, Punjab, India-

147004. Email: [neerajkumar@thapar.edu](mailto:neerajkumar@thapar.edu)

My doctoral advisor, and possesses extensive knowledge in the field of computational fluid dynamics and numerical techniques for biological physics and heat transfer assessment.

---

### **Dr. Pramod Kumar Avti,**

Additional Professor,

Post Graduate Institute of Medical Education and Research (PGIMER), Chandigarh 160012, India.

Email: [avti.pramodkumar@pgimer.edu.in](mailto:avti.pramodkumar@pgimer.edu.in),

My doctoral advisor, and possesses knowledge in the field of nano-bio materials, tissue engineering, regenerative medicine, and biomedical imaging.

---

### **Dr. Bhupendra kumar Chudasama**

Associate Dean, Research and Development Cell

Professor, School of Physics & Materials Science,

Thapar Institute of Engineering and Technology, Patiala, Punjab, India-

147004. Email: [bnchudasama@thapar.edu](mailto:bnchudasama@thapar.edu)

My mentor and Co-Project invigilator specialize in the field of nanomedicine for cancer theranostics and Magnetic Hyperthermia application.

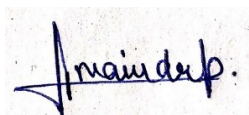
---

## **PERSONAL DETAILS**

Gender / Marital Status: **Male / Married**

Languages Known- **English, Hindi.**

Certified that the information given and being submitted here is true to the best of my knowledge



(Sandeep)

April, 2026